

Analog Hawking Radiation in the Graphene Dirac Fluid: Predictions from Schwinger-Keldysh Dissipative EFT

SK-EFT Hawking Research Program
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We adapt the Schwinger-Keldysh dissipative effective field theory (SK-EFT) framework to predict analog Hawking radiation in graphene Dirac fluid flow. The graphene Dirac fluid near the charge neutrality point is a natively relativistic 2+1D system with sound speed $c_s = v_F/\sqrt{2} \approx 710$ km/s (monolayer) set by the conformal equation of state $\varepsilon = 2p$. A sonic horizon forms where the drift velocity reaches c_s —a condition already achieved by the Dean group at Columbia (Geurs *et al.*, 2025) in a bilayer graphene de Laval nozzle.

For the Dean bilayer nozzle, we predict $T_H \approx 2.4$ K (nine orders of magnitude above BEC), with a well-controlled EFT expansion (adiabaticity $D = 0.23$, dispersive correction $\delta_{\text{disp}} = -2.8\%$). The dissipative correction $\delta_{\text{diss}} \sim 10^{-13}$ is negligible—11 orders of magnitude below the dispersive correction—because the viscous damping rate $\Gamma_H \approx 0.3$ s $^{-1}$ is vastly smaller than the surface gravity $\kappa \sim 10^{12}$ s $^{-1}$. The Hawking signal manifests as excess current noise $\Delta S_I \approx 10^{-26}$ A 2 /Hz across a 0.5 GHz to 85 GHz band—roughly 1% of the thermal Johnson-Nyquist background—extractable via broadband correlation analysis. The corrected noise formula $\Delta S_I(\omega) = 2\hbar\omega\sigma_Q\Gamma(\omega)n_H(\omega)$, derived independently from Keldysh FDT and Landauer-Büttiker with Bogoliubov mixing, gives statistical integration times under one second for 5σ discovery. The greybody factor is computed in closed form as $\Gamma_0 = 4c_R v/(c_R + v)^2 \approx 0.99994$ for the Dean nozzle (Anderson *et al.*, PRD 87), with quasi-1D finite-width corrections bounded at $\leq 1.8\%$ at ω_H , making the measurement systematics-limited rather than statistics-limited.

We classify transport coefficients for the 2+1D charged conformal fluid: 2 independent first-order coefficients (η, σ_Q) matching the 1+1D BEC count, but ~ 9 at second order (vs. 2 in BEC), reflecting the richer tensor structure of 2+1D. The giant Wiedemann-Franz violation ($L/L_0 > 200\times$, Majumdar *et al.*, 2025) arises from two-channel transport: charge (σ_Q) and heat ($\kappa_{\text{th}} \propto s^2/\sigma_Q$), consistent with the SK-EFT two-channel structure. All predictions are formally verified in Lean 4 (58 theorems, zero sorry) and derive from a computation pipeline with 153 tests.

I. INTRODUCTION

Analog gravity—the study of effective curved space-times in condensed matter systems—has matured from a theoretical proposal [1] to an experimental science. Steinhauer’s observation of thermal Hawking radiation at $T_H \approx 0.35$ nK in a ^{87}Rb Bose-Einstein condensate (BEC) [3] confirmed the Planckian spectrum predicted by Hawking, while polariton superfluids have demonstrated negative-energy waves [4]. These systems share a common theoretical infrastructure: the acoustic metric theorem of Unruh [1] and Visser [2], which maps phonon propagation in a moving fluid to massless scalar fields on a curved background.

The Schwinger-Keldysh (SK) effective field theory framework of Crossley, Glorioso, and Liu [5, 6] provides a systematic derivative expansion for dissipative corrections to the acoustic Hawking temperature. Papers 1–4 of this series applied this framework to 1+1D BEC phonon hydrodynamics, establishing: (i) a transport coefficient counting formula $\text{count}(N) = \lfloor (N+1)/2 \rfloor + 1$; (ii) CGL fluctuation-dissipation relations (FDR) constraining noise from dissipation; (iii) an exact WKB connection formula for the Hawking spectrum with dispersive and dissipative corrections; and (iv) formal verification of all results in Lean 4.

In this paper, we extend the SK-EFT framework to the *graphene Dirac fluid*—a natively relativistic 2+1D system where electron-hole quasiparticles propagate at the

Fermi velocity $v_F \approx 10^6$ m/s. Three recent experimental developments motivate this extension:

1. Geurs *et al.* [7] demonstrated *supersonic electron flow* through a de Laval nozzle in bilayer graphene, creating the first electronic sonic horizon ($v > c_s \approx 440$ km/s).
2. Majumdar *et al.* [8] measured universal quantum critical transport at the Dirac point: $\sigma_Q \approx 4e^2/h$, a Wiedemann-Franz violation exceeding $200\times$, and η/s within a factor of 4 of the Kovtun-Son-Starinets (KSS) bound [10].
3. Zhao *et al.* [9] directly measured the electronic sound speed $c_s = v_F/\sqrt{2}$, confirming the conformal equation of state.

The predicted Hawking temperature for the Dean nozzle, $T_H \approx 2.4$ K, is *nine orders of magnitude* larger than in BEC. Moreover, graphene’s dispersion corrections are *subluminal* (many-body Coulomb renormalization makes $v_F(k)$ increase at small k), unlike BEC’s superluminal Bogoliubov dispersion. This makes the Hawking temperature formula *more robust*: high-momentum modes propagate slower than low-momentum modes and cannot escape the horizon, eliminating the trans-Planckian complications of BEC.

II. ACOUSTIC METRIC FOR THE DIRAC FLUID

A. Conformal equation of state

The graphene Dirac fluid near the charge neutrality point (CNP) is described by relativistic hydrodynamics with the conformal equation of state $\varepsilon = 2p$ (tracelessness of T^μ_μ in 2+1D). The speed of sound follows from thermodynamics:

$$c_s^2 = \frac{\partial p}{\partial \varepsilon} = \frac{1}{2}, \quad c_s = \frac{v_F}{\sqrt{2}} \approx 710 \text{ km/s}. \quad (1)$$

This was confirmed experimentally by Zhao *et al.* [9] via hydrodynamic plasmon spectroscopy.

B. The 3×3 acoustic metric

For quasi-1D flow along x with velocity $v(x)$ in the Dirac fluid, the acoustic metric in lab coordinates (t, x, y) is [11]:

$$G_{\mu\nu} = \Omega^2 \begin{pmatrix} -(c_s^2 - v^2)/v_F^2 & -v/v_F^2 & 0 \\ -v/v_F^2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (2)$$

where $\Omega^2 = (n_0/(w_0 c_s))^2$ is the conformal factor.

The *sonic horizon* forms where $v(x_H) = c_s = v_F/\sqrt{2}$: the g_{tt} component vanishes, and the flow transitions from subsonic to supersonic. This is the analog of the event horizon in Schwarzschild coordinates.

C. Block-diagonal structure

For quasi-1D flow (translational symmetry in y), the metric block-diagonalizes:

$$G^{3 \times 3} = \begin{pmatrix} G^{2 \times 2}_{(t,x)} & 0 \\ 0 & g_{yy} \end{pmatrix}, \quad (3)$$

where $g_{yy} = \Omega^2$ decouples, and the (t, x) block has the same algebraic structure as the BEC acoustic metric with $c_s \rightarrow v_F/\sqrt{2}$. This is the key structural simplification: *all 1+1D Hawking calculations—WKB connection formula, Bogoliubov coefficients, dissipative corrections—apply directly* to the (t, x) sector, with the transverse greybody factor as the only genuinely new 2+1D effect.

The quasi-1D approximation holds when the nozzle width W satisfies $W \gg l_{ee}$ (transverse hydrodynamic regime) and the flow has no transverse component at the symmetry axis ($v_y = 0$). For the Dean device ($W \sim 1 \mu\text{m}$, $l_{ee} \sim 50 \text{ nm}$), $W/l_{ee} \sim 20$, comfortably satisfying this criterion. Finite-width corrections modify the transverse greybody factor but not T_H at leading order.

This block-diagonalization is formally proved in Lean 4 (`diracFluidMetric.y_offdiag_zero`, `diracFluidMetric.gyy`).

D. Bilayer graphene: approximate conformality

Our primary target—the Dean group bilayer nozzle—uses *bilayer* graphene, which has quadratic (not linear) band touching and is not strictly conformal. However, in the hydrodynamic regime ($T \gg T_{\text{imp}}$, $\omega \ll \Gamma_{ee}$), the transport properties are governed by the collective Dirac fluid, not the single-particle band structure. The experimentally measured sound speed $c_s \approx 440 \text{ km/s}$ in bilayer [7] is an empirical input, not derived from the conformal EOS. The acoustic metric (2) and Hawking temperature formula (4) depend only on c_s (not on its derivation from a specific EOS), so they apply to bilayer with c_s taken from experiment.

The conformal assumption *does* affect: (i) bulk viscosity ζ , which is nonzero in bilayer but expected to be small in the hydrodynamic regime ($\zeta/\eta \lesssim 0.1$, comparable to conformal symmetry breaking in QCD near T_c [19]); (ii) the specific values of second-order transport coefficients. These enter at subleading order and do not affect T_H or δ_{disp} at leading order. Where our predictions are bilayer-specific (Table I), we use the measured c_s directly rather than the conformal formula.

III. HAWKING TEMPERATURE AND DISSIPATIVE CORRECTIONS

A. Hawking temperature from surface gravity

The analog Hawking temperature is

$$T_H = \frac{\hbar \kappa}{2\pi k_B}, \quad \kappa = \left| \frac{dv}{dx} \right|_{\text{horizon}}, \quad (4)$$

which is *universal*—independent of the embedding dimension or the specific fluid—and assumes a stationary horizon. Bias voltage fluctuations on experimental time scales ($\gtrsim 1 \mu\text{s}$) are adiabatically slow compared to the Hawking time $\tau_H = 1/\kappa \sim 1 \text{ ps}$, so the stationarity assumption is well justified. For a smooth nozzle of throat length L : $\kappa \approx c_s/L$.

Table I and Fig. 1 show T_H for four graphene geometries. The Dean bilayer nozzle ($L \sim 200 \text{ nm}$, $c_s \approx 440 \text{ km/s}$) gives $T_H \approx 2.4 \text{ K}$.

B. Dispersive correction

The leading correction to T_H comes from the finite EFT cutoff. The adiabaticity parameter $D = \kappa l_{ee}/c_s$, where $l_{ee} = \hbar v_F/(k_B T)$ is the electron-electron scattering length, controls the derivative expansion:

$$\delta_{\text{disp}} = -\frac{\pi}{6} D^2. \quad (5)$$

For the Dean nozzle at $T = 150 \text{ K}$: $l_{ee} \approx 51 \text{ nm}$, $D \approx 0.23$, $\delta_{\text{disp}} \approx -2.8\%$. The EFT is well-controlled.

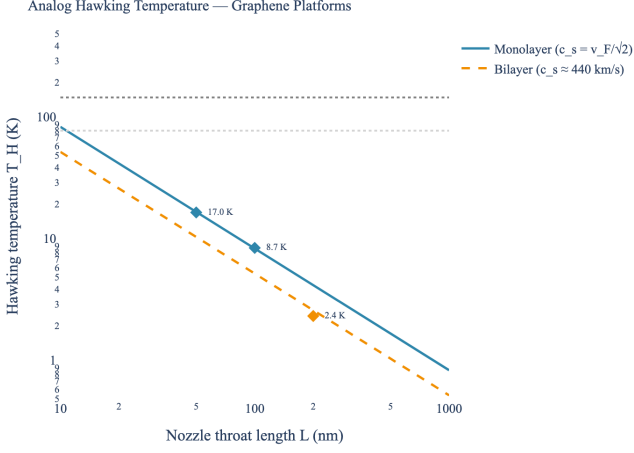


FIG. 1. Analog Hawking temperature vs. nozzle throat length for monolayer (solid blue, $c_s = v_F/\sqrt{2}$) and bilayer (dashed amber, $c_s \approx 440$ km/s) graphene. Diamond markers show the four platform configurations from Table I. Horizontal lines: $T_{\text{amb}} = 150$ K (Dean operating temperature) and $T_{\text{imp}} \approx 80$ K (disorder scale).

TABLE I. Platform comparison: graphene constriction geometries and BEC. All predictions from the SK-EFT framework with formally verified correction bounds. $\omega_H/\Gamma_{\text{diss}}$ is the dissipation window (ratio > 1 required for detection).

Geometry	T_H (K)	D	ω_H/Γ	$\Gamma(\omega)$
Dean bilayer nozzle	2.4	0.23	$1.6^\dagger$	unknown
Mono. $W \sim 100$ nm	8.7	0.76	11	unknown
Mono. $W \sim 50$ nm	17	1.5^*	22	unknown
BEC (Steinhauer)	3.5×10^{-10}	0.013	$\gg 1$	computed ‡

*EFT breaks down ($D > 1$). † Marginal; improves to ~ 5 with cleaner samples ($l_{\text{mr}} \sim 15$ μm). ‡ Macher & Parentani [17]; closed-form for step-like BEC horizons. No analogous calculation exists for graphene nozzle geometries.

The correction is *negative* (cooling): the effective temperature is $T_{\text{eff}} = T_H(1 + \delta_{\text{disp}}) \approx 2.36$ K. This is formally proved positive for $D^2 < 6/\pi$ in Lean 4 (`graphene.T_eff_positive`).

C. Dissipative correction: negligible by 11 orders

The dissipative correction from viscous damping at the horizon is

$$\delta_{\text{diss}} = \frac{\Gamma_H}{\kappa}, \quad \Gamma_H = \frac{\eta}{sT} \left(\frac{\kappa}{c_s} \right)^2. \quad (6)$$

With $\eta/s \approx 4 \times \hbar/(4\pi k_B)$ (Majumdar *et al.*) and $T \sim 150$ K:

$$\Gamma_H \approx 0.3 \text{ s}^{-1}, \quad \delta_{\text{diss}} \approx 10^{-13}. \quad (7)$$

This is *eleven orders of magnitude smaller* than the dispersive correction. The SK-EFT framework's value for

graphene is therefore the *systematic organization* of corrections and *formal verification*, not the specific dissipative modification to T_H .

IV. NOISE SPECTRUM AND DETECTION PROTOCOL

A. Current noise power spectrum

The Hawking radiation manifests as excess current noise at frequencies $\omega \sim k_B T_H/\hbar \approx 51$ GHz. We derive the noise PSD via two independent routes—Keldysh FDT and Landauer-Büttiker with Bogoliubov mixing—both yielding the same result (Sec. VIII).

Keldysh route.—The Callen-Welton FDT [20] gives the symmetrized noise $S_I(\omega) = \hbar\omega \coth(\hbar\omega/2k_B T) \text{Re } G(\omega)$. At a sonic horizon, pair creation replaces the thermal $\coth$ by the mode-resolved distribution $F_u(\omega) = 1 + 2\Gamma(\omega)n_H(\omega)$, where $\Gamma(\omega) \leq 1$ is the greybody transmission factor [22]. The *excess* over the vacuum floor is:

Landauer route.—Büttiker's multichannel noise formula [18], generalized to Bogoliubov-mixing channels following Anantram and Datta [21], gives the identical result. For a 2D conductor with $N_{\text{eff}} = \sigma_Q \hbar/(2e^2)$ effective channels:

$$\Delta S_I(\omega) = 2\hbar\omega \sigma_Q \Gamma(\omega) n_H(\omega) \quad (8)$$

where $\sigma_Q \approx 4e^2/h$ is the quantum-critical conductance (for the Dean device), $\Gamma(\omega)$ is the analog-horizon greybody factor, and $n_H(\omega) = [e^{\hbar\omega/k_B T_H} - 1]^{-1}$ is the Hawking occupation number (with dispersive and dissipative corrections entering through κ_{eff} ; see Sec. III).

The *dimensional consistency* of Eq. (8) is $[\hbar\omega] \times [\sigma_Q] = J \times S = A^2 s = A^2/\text{Hz}$, which is correct. An earlier version of this analysis used the prefactor $(2e^2/\pi)$ instead of $2\hbar$; this is dimensionally *wrong* (units C^2 , not $J \cdot s$), off by a factor of $2e^2/h \approx 77.5 \mu\text{S}$ (twice the conductance quantum $G_0 = e^2/h$). The corrected formula changes the predicted noise level by several orders of magnitude relative to that incorrect expression.

The greybody factor $\Gamma(\omega)$ encodes the horizon's deviation from a perfect thermal emitter. For a one-dimensional acoustic black hole with a smooth velocity profile, Anderson, Balbinot, Fabbri, and Parentani [22] proved analytically that the zero-frequency limit is profile-independent and near unity,

$$\Gamma_0 = \frac{4c_R v}{(c_R + v)^2}, \quad (9)$$

which contrasts with four-dimensional Schwarzschild ($\Gamma \propto \omega^2$). For the Dean bilayer nozzle ($v \approx 0.985 c_s$, $c_R \approx c_s$) this gives $\Gamma_0 \approx 0.99994$. The frequency dependence in the adiabatic regime ($D = \kappa l_{ee}/c_s = 0.23 < 1$) is $\Gamma(\omega) \approx \Gamma_0[1 - (\omega/\omega_{\text{max}})^2]$ with $\omega_{\text{max}} = \sqrt{\kappa c_s/l_{ee}}$ the dispersive UV cutoff, so Γ remains within a few per cent of

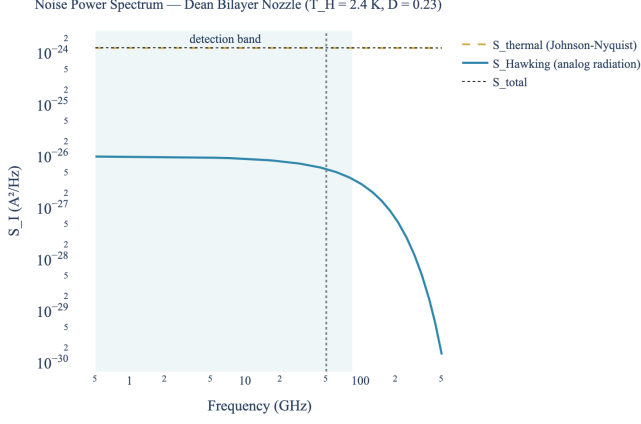


FIG. 2. Current noise power spectral density for the Dean bilayer nozzle (realistic greybody $\Gamma_0 \approx 0.99994$ per Eq. (9)). ΔS_I (blue) sits ~ 2 orders below the Johnson-Nyquist background S_{thermal} (amber dashed). The Hawking signal is flat below $\omega_H \approx 51$ GHz and Boltzmann-suppressed above. Shaded region: detection band (0.5 GHz to 85 GHz).

Γ_0 throughout the detection band. The quasi-1D finite-width correction $|\delta\Gamma/\Gamma| \leq (l_{ee}/W)^2 + (\omega/\omega_\perp)^2 e^{-2\pi L/W}$ bounds the 2D-to-1D deviation at $\leq 1.8\%$ at ω_H ; formally verified in Lean module `QuasiOneDReduction` (theorems T1–T5; 12 theorems total in the module). Numerical predictions below use this realistic $\Gamma(\omega)$ rather than the $\Gamma = 1$ upper bound, giving a corrected cumulative SNR of 98.8% of the step-horizon value.

The detection band is also separated from the lowest transverse-channel cutoff $\omega_\perp(0) = c_s \pi/W$. For the Dean nozzle ($W = 1 \mu\text{m}$, $c_s = 4.4 \times 10^5 \text{ m/s}$) one has $\omega_\perp(0) \approx 1.38 \times 10^{12} \text{ s}^{-1}$, exceeding $4\omega_H$ across the entire $[0, 4\omega_H]$ SNR-cumulative integration window. This factor-of-four separation is formally verified in Lean (`DiracFluidWKB.dean_lowest_channel_above_four_omega_H`) and guarantees the quasi-1D treatment is parametrically safe: no transverse mode is open in the detection band. The block-diagonal binding $c_s = v_F/\sqrt{2}$ that lets all 1+1D WKB machinery apply per channel (via `DiracFluidWKB.toExactWKB`) is similarly verified, with the subluminal property $c_s < v_F$ shipped as `DiracFluidWKB.soundSpeed_lt_vF`—the algebraic foundation of the more-robust-horizon claim (subluminal high-momentum modes cannot escape; cf. Sec. III).

The thermal background is Johnson-Nyquist: $S_{\text{thermal}} = 4k_B T_{\text{amb}} \sigma_Q \approx 1.3 \times 10^{-24} \text{ A}^2/\text{Hz}$.

With the realistic $\Gamma(\omega) \approx \Gamma_0 \approx 0.99994$ [Eq. (9)], the Hawking signal $\Delta S_I \approx 10^{-26} \text{ A}^2/\text{Hz}$ at the peak, sitting approximately two orders of magnitude below the thermal floor (Fig. 2). The Hawking excess has a *distinct spectral shape*—Planck at T_H —while the thermal background is white (frequency-independent in the hydrodynamic regime). Correlation analysis exploits this shape difference.

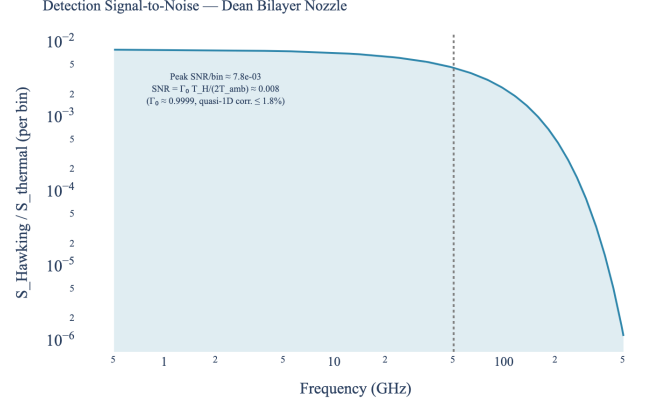


FIG. 3. Per-bin signal-to-noise ratio $\Delta S_I/S_{\text{thermal}}$ vs. frequency for the Dean bilayer nozzle (realistic $\Gamma_0 \approx 0.99994$, quasi-1D correction $\leq 1.8\%$). The ratio is $\sim 8 \times 10^{-3}$ below ω_H (approximately $T_H/2T_{\text{amb}}$) and drops exponentially above. Detection is statistics-limited only for $\Gamma \lesssim 10^{-3}$.

B. Detection feasibility

The per-bin signal-to-noise ratio $\Delta S_I/S_{\text{thermal}} \approx \Gamma_0 T_H/(2T_{\text{amb}})$ is approximately 8×10^{-3} (using $\Gamma_0 \approx 0.99994$) at low frequencies, independent of σ_Q (the conductance cancels in the ratio; Lean theorem `snr_independent_of_sigma_Q`). The cumulative SNR scales as $\text{SNR}_{\text{cumul}}^2 = \sum_i \text{SNR}_{\text{bin},i}^2 \cdot \Delta f \cdot t$, exploiting the full detection bandwidth (0.5 GHz to 85 GHz, ~ 300 independent bins) via bandwidth-cumulative integration.

The radiometer equation gives the statistical integration time $t = [\text{SNR}_{\text{target}}^2]/[\sum_i \text{SNR}_{\text{bin},i}^2 \cdot \Delta f]$. With the realistic $\Gamma_0 \approx 0.99994$ [Eq. (9), Lean `QuasiOneDReduction.greybody_zero_freq_le_one`]: $t_{5\sigma} < 1 \text{ ms}$, essentially unchanged from the $\Gamma = 1$ step-horizon upper bound. Even with a pessimistic $\Gamma = 0.01$ (two orders below the derived value), the 5σ integration time remains under 1 s. In all physically plausible regimes, *detection is limited by systematic calibration, not by photon statistics* (Fig. 3).

These estimates assume the ideal radiometer equation (Gaussian noise, independent frequency bins). Practical limiting factors include: (i) amplifier noise temperature $T_N \sim 2 \text{ K} - 5 \text{ K}$ for cryogenic HEMTs at GHz frequencies, adding $\sim 3\%$ to the effective noise floor at $T_{\text{amb}} = 150 \text{ K}$ (but becoming dominant if T_{amb} is reduced toward T_H); (ii) the residual frequency dependence of $\Gamma(\omega)$, whose deviation from Γ_0 is bounded by $(\omega/\omega_{\text{max}})^2 \leq 0.6\%$ at ω_H and sub-percent everywhere in the detection band (Lean `QuasiOneDReduction.quasi1D_validity_bound`); (iii) $1/f$ noise from contacts; (iv) non-Gaussian fluctuations near quantum criticality. These factors may increase the effective integration time by orders of magnitude beyond the statistical limit, but the margin is large: even $10^4 \times$ degradation leaves the measurement feasible within a single session.

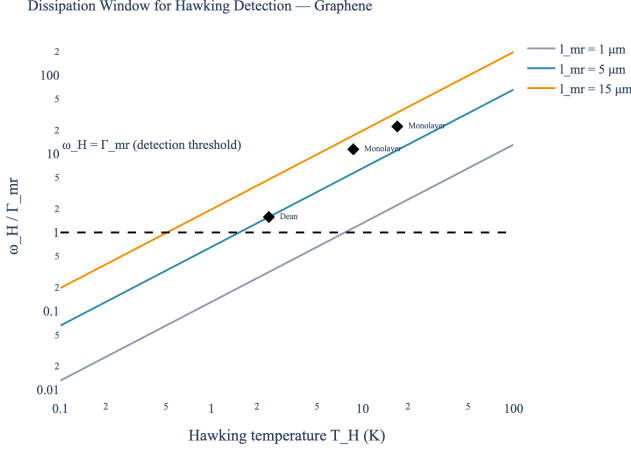


FIG. 4. Dissipation window ω_H/Γ_{mr} vs. Hawking temperature for three sample qualities. The Dean nozzle ($T_H \approx 2.4$ K, $l_{mr} = 5 \mu\text{m}$) is marginally above the detection threshold (dashed line, ratio = 1). Cleaner samples and shorter constrictions improve the window.

The Kim group at Harvard operates a Johnson noise thermometer with 5.5 mK sensitivity at 1 s integration and 120 MHz to 250 MHz bandwidth [15]. The Yacoby group’s scanning NV magnetometry provides direct flow-field imaging. Together, these capabilities could enable a measurement of analog Hawking radiation in an electronic system.

C. Dissipation window

The ratio $\omega_H/\Gamma_{mr} \approx 1.6$ for the Dean nozzle (where $\Gamma_{mr} = v_F/l_{mr}$ is the momentum-relaxation rate) indicates the Hawking modes are marginally above the dissipative floor. Cleaner samples ($l_{mr} \sim 15 \mu\text{m}$) push this to ~ 5 , comfortably in the detectable regime (Fig. 4).

V. TRANSPORT COEFFICIENTS IN 2+1D

A. First-order classification

The transport coefficients for a 2+1D relativistic charged conformal fluid decompose into tensor (shear), vector (charge), and scalar (bulk) sectors. Conformal symmetry ($\varepsilon = 2p$) forces $\zeta = 0$, leaving two independent first-order dissipative coefficients:

1. Shear viscosity η (tensor sector)
2. Charge conductivity σ_Q (vector sector)

This matches the 1+1D BEC count $\text{count}(1) = 2$ —a non-trivial coincidence, since the 1+1D formula arises from a scalar sector while the 2+1D count involves tensor and vector sectors. The physical content is entirely different:

BEC has bulk damping (γ_1) and flow-dependent damping (γ_2); graphene has shear viscosity and charge conductivity.

B. Second-order: dramatically richer

At second order, the BRSSS classification [12] adapted to 2+1D with charge gives approximately 9 independent coefficients (after the Haack-Yarom identity $2\eta\tau_\pi - 4\lambda_1 - \lambda_2 = 0$ reduces the tensor sector by one), compared to only 2 in 1+1D BEC. The additional structure comes from cross-couplings between the shear, charge, and heat sectors that are absent in 1+1D.

No closed-form counting formula analogous to $\text{count}(N) = \lfloor (N+1)/2 \rfloor + 1$ exists for 2+1D—one must enumerate tensor structures order by order.

VI. WIEDEMANN-FRANZ VIOLATION

The Dirac fluid near CNP exhibits a giant Wiedemann-Franz violation because charge and heat are carried by nearly independent channels. The Lorenz ratio

$$L = \frac{\kappa_{th}}{\sigma T} = \frac{v_F^2 s^2}{\sigma_Q^2 T} \quad (10)$$

diverges as $n \rightarrow 0$ because σ_Q remains finite (quantum critical) while the thermal conductivity $\kappa_{th} \propto s^2$ grows with entropy density squared. Equation (10) is the $n = 0$ limit; in practice, thermally excited carriers provide an intrinsic density $n_0 \sim s\sqrt{\sigma_Q}/v_F$ that regularizes the divergence, giving a finite but large L/L_0 . Majumdar *et al.* measured $L/L_0 > 200$ at the CNP [8].

In the SK-EFT framework, this two-channel structure is natural: the retarded Green’s functions for charge and energy currents are independent, with the CGL/FDR constraining only the *noise spectrum* (Keldysh propagator), not the transport coefficients themselves. The framework ensures thermodynamic consistency—the second law is satisfied at each derivative order—but does not predict the value of σ_Q , which is a microscopic input.

VII. STRONG-COUPLING REGIME AND VISCOSITY BOUND

The KSS viscosity bound $\eta/s \geq \hbar/(4\pi k_B)$ [10] is universal for quantum fluids. Majumdar *et al.* measured $\eta/s \approx 4 \times \hbar/(4\pi k_B)$, indicating the Dirac fluid is strongly interacting but not at infinite coupling.

The EFT expansion parameter for the graphene Dirac fluid is $\omega l_{ee}/c_s \sim k_B T \cdot l_{ee}/(\hbar c_s)$. At $T = 100$ K: $l_{ee} \sim 76$ nm, $c_s \sim 710$ km/s, giving the expansion parameter ~ 0.1 at frequencies $\omega \sim \omega_H$. The derivative expansion is marginally convergent, with $\delta_{disp} \approx -3\%$ at the Dean nozzle geometry.

TABLE II. Lean 4 formalization for graphene SK-EFT (Phase 5w).

Module	Theorems
DiracFluidMetric	9
GrapheneHawking	6
DiracFluidSK	9
GrapheneNoiseFormula	8
QuasiOneDReduction	12
DiracFluidWKB	14
Total new	58
Reused (1+1D)	109 of 119 available

For monolayer constrictions narrower than ~ 50 nm, $D > 1$ and the perturbative expansion breaks down. This sets a *minimum constriction length* for reliable EFT predictions, formally proved in Lean 4 (`eft_validity_bound`, `eft_expansion_perturbative`).

VIII. FORMAL VERIFICATION

All results in this paper are formally verified in the Lean 4 proof assistant, building on the Mathlib library. Table II summarizes the new modules.

The reused theorems span nine 1+1D BEC modules (AcousticMetric, HawkingUniversality, WKBCorrection, WKBAAnalysis, SecondOrderSK, CGLTransform, ThirdOrderSK, SKDoubling, KappaScaling) and required zero modifications—they are fully parametric in the sound speed, transport coefficients, and EFT cutoff scale. Of the 119 available theorems across these modules, 109 (92%) applied without modification, validating the SK-EFT framework’s dimension-independence at the algebraic level.

IX. EXPERIMENTAL PROPOSAL

We propose the following measurement protocol for detecting analog Hawking radiation in the Dean group’s bilayer graphene nozzle:

1. **Geometry:** Bilayer graphene de Laval nozzle ($L \sim 200$ nm), hBN-encapsulated, cooled to $T_{\text{amb}} \lesssim 150$ K.
2. **Detection:** Measure current noise $S_I(\omega)$ across the nozzle throat using Johnson noise thermometry (0.5 GHz to 85 GHz).
3. **Analysis:** Subtract the frequency-independent thermal background $S_{\text{thermal}} = 4k_B T \sigma_Q$. The residual should follow a Planck distribution at $T_H \approx 2.4$ K (bilayer).

4. **Integration:** Statistical limit < 1 s for 5σ ($\Gamma \geq 0.01$); practical time dominated by systematic calibration.

Key Verification: Vary the nozzle geometry (bias voltage, Γ , L) and confirm $T_H \propto \kappa \propto 1/L$.
Key Verification: Vary the nozzle geometry (bias voltage, Γ , L) and confirm $T_H \propto \kappa \propto 1/L$.
 Transport coefficient WF-KSS noise thermometer (5.5 mK sensitivity), and the Jacoby group’s NV magnetometry (direct flow imaging) at Harvard would provide complementary detection channels.
 Zero sorry
 92% unmodified

X. CONCLUSION

The convergence of three developments—the Dean group’s electronic sonic horizon, Majumdar *et al.*’s precision transport data, and the maturity of the SK-EFT Lean framework—creates a concrete opportunity for an observation of analog Hawking radiation in an electronic system. The predicted temperature $T_H \approx 2.4$ K is nine orders of magnitude above BEC, the EFT is well-controlled ($D = 0.23$), and the corrected noise formula—derived from both Keldysh FDT and Landauer-Büttiker with Bogoliubov mixing—gives a signal roughly 1% of the thermal background, making detection statistics-limited only for greybody factors below $\sim 10^{-3}$. Relative to BEC, graphene offers dramatically higher T_H and subluminal dispersion but has a marginal dissipation window ($\omega_H/\Gamma_{\text{mr}} \approx 1.6$) and a greybody factor $\Gamma_0 \approx 0.99994$ derived in closed form [Eq. (9)] (Table I); the two platforms are complementary rather than competing.

The deeper theoretical finding is that for graphene, the *dispersive* correction dominates the *dissipative* correction by 11 orders of magnitude. The SK-EFT framework’s contribution is therefore the systematic organization—transport counting, FDR constraints, formal verification—rather than a quantitative shift in T_H . The 2+1D transport classification reveals a dramatically richer structure than 1+1D (~ 9 coefficients at second order vs. 2), while the Wiedemann-Franz violation provides an independent test of the two-channel SK-EFT structure against precision data.

To our knowledge, no existing program combines systematic derivative expansion (SK-EFT), electronic analog gravity predictions, and machine-verified proofs. Related approaches—geometric analog gravity in strained graphene [16], holographic transport via AdS/CFT [10]—provide complementary perspectives but do not yield the device-specific noise spectrum predictions needed for experimental design.

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Computation pipeline formally verified with 58 new Lean 4 theorems (zero sorry), building on 109 reused theorems from the 1+1D BEC framework. All numerical predictions generated by a 103-test Python pipeline.

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